

ENTERPRISE ERP IMPLEMENTATION ROADMAP & PAYMENT GOVERNANCE

Module 4: Sales Process, Counter & Handheld POS, Multi-Branch Billing, Sales Return & Branch Financial Control

Comprehensive milestone agreement, engineering effort breakdown, resource cost schedule, and acceptance criteria based on the verified Business Analysis Report **DOC-004 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

DOCUMENT ID

SL-POP-ERP-MS-004

BA REFERENCE DOCUMENT

DOC-004 v1.0 (30-Mar-2026)

TOTAL ENGINEERING EFFORT

15 Working Weeks (3.75 Months)

FIXED PROJECT COST

BD 5,113.636

TARGET ARCHITECTURE

Multi-Branch Cloud + Offline
POS

DOCUMENT STATUS

FINAL PROPOSAL

Executive Summary & Master ERP Context

The **Sales Process & Branch Financial Control Module** (Module 4) constitutes the high-velocity operational revenue engine of the Popular Auto Spare & A/C Parts ERP platform. Auto spare parts distribution demands rapid item identification across hundreds of thousands of active SKUs, instant compatibility cross-referencing, multi-tier pricing governance, flexible walk-in and credit invoicing, and mobile handheld scanner assistance on the showroom floor.



Core Objective & Business Purpose

To establish an ultra-responsive, fault-tolerant sales ecosystem connecting floor sales representatives using handheld Android barcode/P-Code scanners with desktop billing counters, incorporating multi-level discount authorization, automated credit limit enforcement, unified sales returns, and seamless double-entry branch accounting.

Master ERP Transformation Architecture (9 Core Process Modules)

CURRENT SCOPE: MODULE 04
ACTIVE

- 01 PCode Gen & Item Master
- 02 Vendor & Purchase Flow
- 03 Stock Verification & Location
- 04 Sales & POS Checkout
- 05 Finance, Tax & VAT Accounting
- 06 Administration & Security
- 07 HRMS & Biometric Payroll
- 08 Hardware & Barcode Infrastructure



Handheld Floor POS & QR Handoff

Floor salesmen build digital carts via P-Code scanning, real-time stock lookup, and generate encrypted QR handoff tokens for instant 1-scan checkout at cashier counters.

Android PDA

QR Cart Transfer

P-Code Scan



Intelligent Pricing & Search

4-Tier search (P-Code, OEM, Cross-Ref, Vehicle Make/Model), tiered wholesale/retail pricing rules, and multi-level manager discount approval thresholds.

Smart Lookup

Tiered Margin

Approval Matrix



Branch Accounting & Day Close

Full suite of branch accounting vouchers (CRV, CPV, JV, Contra, Petty Cash), automated double-entry GL posting, and physical cash drawer reconciliation with Day-Close lock.

Branch Vouchers

Petty Cash

EOD Drawer Lock

Module 4: Functional Scope & Architecture

Breakdown (BA Ref: DOC-004)

Derived directly from the 58 pages of **DOC-004: Sales Process Business Analysis Report**, the functional scope is segmented into 10 key architectural domains:

2.1 Customer Master & Multi-Branch Governance

- **Single Source of Truth:** Centralized customer database preventing multi-branch duplicate records while preserving branch-origin ownership.
- **Customer Classifications:** Dedicated profiles for Retail, Corporate Accounts, Garages/Workshops, Wholesalers, and Credit Customers.
- **Dual Credit Controls:** Branch credit limit & enterprise credit ceiling with automated overdue day locks and manager override workflows.
- **Settlement Ledger:** Multi-currency customer ledgers, consolidated statement of accounts (SOA), and automated aging analysis.

2.2 Intelligent Item Search & Pricing Governance

- **Multi-Parameter Search:** Search by P-Code barcode, OEM Part No., Supplier Part No., Part Name, and Vehicle Compatibility (Make, Model, Year, Engine).
- **Multi-Branch Stock Matrix:** Real-time visibility into local branch rack/bin locations and cross-branch warehouse availability.
- **Tiered Pricing Engine:** Base retail price, garage discount tier, wholesale tier, and customer-specific negotiated rates.
- **Approval Threshold Matrix:** Salesman allowed discount (0-5%), Branch Manager approval (5-15%), and General Manager override (>15%).

2.3 Handheld Mobile POS & QR Cart Handoff

- **Floor Sales Mobility:** Android handheld PDA app for rapid shelf scanning, real-time item validation, and temporary stock reservation.

- **QR Cart Handoff:** Generates encrypted dynamic QR code on salesman's screen; cashier scans once to load cart instantly into counter POS.
- **Multi-Salesperson Attribution:** Tracks floor sales rep, assisting rep, and cashier with automated commission percentage splitting.
- **Exception Handling:** Cart on-hold/resume, cart merging, scan failure fallbacks, and automatic stock release on cancellation.

2.4 Sales Document Modules Suite

- **Quotation Module:** Multi-currency quoting, price validity periods, revision versioning, and 1-click conversion to Delivery Note or Tax Invoice.
- **Delivery Note (DN):** Direct DN or Quotation-based DN, partial dispatch tracking, and pending-invoice reconciliation.
- **VAT Tax Invoice:** Full compliance with Bahrain/GCC VAT regulations, QR e-invoicing, multi-mode payment split (Cash, Card, BenefitPay, Cheque, Credit).
- **Cash Memo:** Fast-lane retail POS receipt generation with instant thermal printing and cash drawer trigger.

2.5 Unified Sales Return & Restocking Engine

- **Triple Return Workflows:** Invoice-referenced return, Delivery Note return, and Direct/Unbilled return (manager authorization required).
- **Item Condition Grading:** Restockable Good Condition (instant P-Code return to bin) vs Damaged/Defective (quarantine location & vendor RMA).
- **Financial Reversals:** Credit Note generation, cash refund or customer credit balance adjustment, and tax VAT reversal.
- **Commission Clawback:** Automated deduction/adjustment of sales commission from involved sales representatives.

2.6 Branch Accounting & Voucher Management

- **Complete Voucher Suite:** Cash Receipt Voucher (CRV), Cash Payment Voucher (CPV), Journal Voucher (JV), Contra Voucher (CV), Expense Voucher (EV).
- **Petty Cash Management:** Imprest float tracking, daily operational expense vouchers, receipt attachments, and replenishment requests.

- **Standardized Numbering:** Branch-coded sequential document prefixes ('BR01-CRV-2026-XXXX').
- **Financial GL Posting:** Real-time automated double-entry ledger posting to Chart of Accounts (COA).

2.7 Less Item Handling & Branch-to-Garage Sales

- **Less Item Handling (DOC-002 Link):** Sales counter logging of stock shortages and customer demand lost sales, automated feed into procurement planning.
- **Branch-to-Garage Sales:** Dedicated workshop sales portal, vehicle registration tracking, recurring credit settlement, and mechanic commission logging.
- **Branch Transfers (DOC-003 Link):** Inter-branch stock requests triggered directly from counter POS when local stock is depleted.

2.8 Daily Closing, Cash Drawer & Audit Lock

- **End-of-Day (EOD) Drawer Count:** Physical currency breakdown vs ERP system expected balance with automated variance (shortage/overage) logging.
- **Payment Gateway Reconciliation:** Card terminal / BenefitPay batch totals verification and settlement slip attachment.
- **Manager Sign-off & System Lock:** Mandatory digital sign-off by Branch Manager; irreversible transactional lock preventing backdated modifications.
- **Cash Settlement to Main:** Physical cash transfer tracking to Head Office / Bank Deposit with dual-custody verification.

Project Team Structure & Dedicated Resource Pricing

(15 Weeks / 3.75 Months)

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 15-week (3.75 months / 75 working days) delivery of Module 4 is structured as follows (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Solution Architect	BD 545.455	BD 136.364	BD 3.409	3.75 Months (15 Wks)	BD 2,045.456
Back-End Lead Engineer (PHP MVC / REST APIs)	BD 227.273	BD 56.818	BD 1.420	3.75 Months (15 Wks)	BD 852.274
Front-End Lead Engineer (Mobile POS / React UI)	BD 227.273	BD 56.818	BD 1.420	3.75 Months (15 Wks)	BD 852.274
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.75 Months (15 Wks)	BD 767.044
QA & Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	3.75 Months (15 Wks)	BD 596.591

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Total Dedicated Engineering Team	BD 1,363.636 / mo	BD 340.909 / wk	BD 8.523 / hr	15 Weeks (3.75 Mo)	BD 5,113.636



Fixed Price Guarantee

Total project investment for Module 4 is fixed at **BD 5,113.636**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

Detailed Milestone Breakdown & Deliverables (15-Week Roadmap)

The implementation is organized into 5 sequential 3-week sprint milestones, guaranteeing rapid delivery, continuous testing, and transparent progress tracking:

1

Milestone 1: Customer Master Governance, Credit Matrix & Intelligent Search

WEEKS 1 – 3

Setup of centralized Customer Master, multi-branch classification, dual credit limit control engine, overdue freeze logic, and the 4-tier intelligent item search engine with real-time multi-branch stock matrix.



Customer Hierarchy & SOA

Centralized database, branch assignment, credit period (30/60/90 days), and ledger.



Intelligent Search Engine

P-Code, OEM part no, cross-reference, and vehicle make/model compatibility lookup.



Pricing & Margin Matrix

Wholesale, retail, garage discount tiers, and salesman discount limit controls.

2

Milestone 2: Handheld Mobile POS App, Scanner Cart & QR Handoff

WEEKS 4 – 6

Development of mobile Android PDA application for showroom salesmen, rapid P-Code barcode scanning, real-time stock reservation, encrypted QR code cart generation, and 1-scan cashier handoff.



Mobile Scanner UI

Touch-optimized Android interface, offline-tolerant queue, and fast SKU scanning.

3

✓ **QR Cart Transfer Protocol**

Secure dynamic QR encryption, counter POS scan receiver, and cart merge/resume.

✓ **Cart Exception Workflows**

Cart On-Hold, partial item cancellation, device swap, and timeout stock release.

Milestone 3: Sales Documents Suite, Multi-Salesperson Attribution & Garage Sales

WEEKS 7
– 9

Engineering of Quotations, Delivery Notes (DN), GCC VAT-compliant Tax Invoices, Cash Memos, multi-mode payment split, multi-salesperson commission attribution, and dedicated garage sales portal.

✓ **Full Document Lifecycle**

Quote → DN → Invoice conversion with version history and pending invoice tracking.

✓ **Multi-Salesperson Split**

Lead owner, floor sales rep, and cashier attribution with target tracking.

✓ **Garage & Workshop Flow**

Dedicated garage billing, vehicle registration tagging, and periodic statement dispatch.

4

Milestone 4: Unified Sales Return Engine, Restock / Quarantine & Commission Clawback

WEEKS 10
– 12

Implementation of invoice-linked and direct returns, physical condition grading (Good vs Damaged), automated Credit Note issuance, customer ledger/VAT adjustments, and commission clawback.

✓ **Unified Return Engine**

Invoice-based, DN-based, and direct unbilled returns with manager approval workflow.

✓ **Condition Inspection & PCode Reversal**

Restocking to shelf rack/bin vs quarantine holding for supplier warranty RMA.

✓ **Financial & Commission Adjustment**

Credit Notes, customer balance refunds, VAT tax adjustment, and commission clawbacks.

Milestone 5: Branch Accounting Vouchers, EOD Cash Drawer Closing & Consolidated GL

WEEKS 13
– 15

Rollout of branch vouchers (CRV, CPV, JV, Contra, Expense), Petty Cash management, End-of-Day physical cash drawer count reconciliation, manager sign-off, system lock, and consolidated GL posting.

✓ **Branch Voucher Suite**

CRV, CPV, JV, CV, EV with sequential numbering and double-entry GL automation.

✓ **EOD Drawer Count & Lock**

Cash denomination count, POS terminal settlement, variance log, and day-closing lock.

✓ **Main Cash Settlement & UAT**

Cash transfer to Head Office/Bank, inter-branch reconciliation, and final multi-branch UAT sign-off.

Milestone Payment Structure & Invoicing Schedule (15-Week Model)

Payments are structured in 5 equal milestone disbursements of **20.00% (BD 1,022.727 each)**, tied strictly to formal review and milestone acceptance:

Milestone Phase	Deliverable Description	Timeline	Percentage	Amount (BHD)	Payment Trigger & Acceptance Criteria
Milestone 1	Customer Master Governance, Credit Matrix & Intelligent Search	Weeks 1–3	20.00%	BD 1,022.727	Demonstration of Centralized Customer Master, Credit Limit Engine & 4-Tier Search
Milestone 2	Handheld Mobile POS App, Scanner Cart & QR Handoff	Weeks 4–6	20.00%	BD 1,022.727	Live demonstration of Mobile POS scanner cart creation & scan QR cashier handoff
Milestone 3	Sales Documents Suite, Multi-Salesperson Split & Garage Portal	Weeks 7–9	20.00%	BD 1,022.727	Delivery of Quotations, VAT Invoices, Cash Memo and Commission Attribution Engine
Milestone 4	Unified Sales Return Engine,	Weeks 10–12	20.00%	BD 1,022.727	Acceptance of Return Workflows

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
	Restock/Quarantine & Reversals				Credit Note Generation Automated Commission Clawback
Milestone 5	Branch Vouchers, Petty Cash, EOD Drawer Closing & Final GL UAT	Weeks 13-15	20.00%	BD 1,022.727	Verification EOD Cash Drawer Lock GL Posting Main Settlement Final Multi-Branch Sign-off
Total Milestone Project Cost	Complete Module 4 ERP Scope Delivery	15 Weeks	100.00%	BD 5,113.636	Formal Acceptance & Sign-off

Working Calendar, SLA, 15-Day Grace Period & Legal Terms

Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled every alternating Thursday.

15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify deliverables. Feedback submitted within this period is incorporated prior to milestone invoice closure.

SLA & Defect Severity Matrix

Critical POS/Billing Blockers (Severity 1) resolved within 4 hours; Major Functional Defects (Severity 2) resolved within 24 hours; Minor UI/Display inquiries (Severity 3) within 3 business days.

Scope Control & Change Requests

Any feature request exceeding the verified 58-page specifications in DOC-004 will be documented as a formal Change Request (CR) and evaluated based on the SaNDS Lab standard rate card.

Stakeholder Authorization & Sign-Off

By digitally endorsing this Milestone & Payment Structure document, all parties confirm adherence to the deliverables, 15-week timeline, and financial schedule defined for **SL-POP-ERP-MS-004**:

DEVELOPER / ARCHITECT

Ancy Varghese Thekkan
Software Architect, SaNDS Lab

✓ PREPARED (27-MAR-2026)

COMPLETED

VERIFIED BY (DEVELOPER)

Ajit Kumar KV
CEO & Managing Director, SaNDS Lab

✓ VERIFIED (30-MAR-2026)

CHECKED & VERIFIED

CONSULTANT REVIEW

UniGlobal Business Solutions
Project Management Consultant

🕒 PENDING REVIEW

IN REVIEW

CLIENT APPROVAL

Popular Auto Spare & A/C Parts
Authorized Executive Signatory

🕒 PENDING APPROVAL

PENDING SIGNATURE